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WAN Technology

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Objectives

- WAN Technologies Overview
- WAN Technologies
- WAN Design

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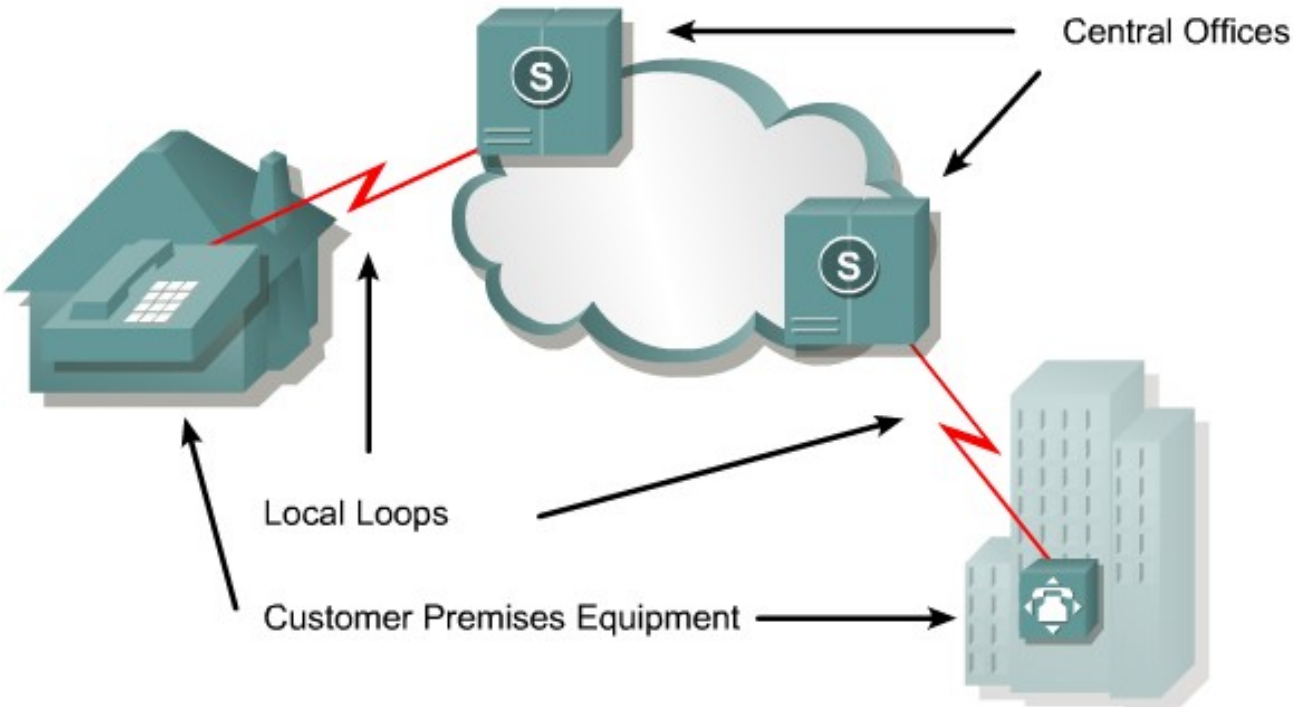
WAN Technology

A WAN uses data links provided by carrier services to access the Internet and connect the locations of an organization to each other, to locations of other organizations, to external services, and to remote users.

WANs generally carry a variety of traffic types, such as voice, data, and video.

Telephone and data services are the most commonly used WAN services.



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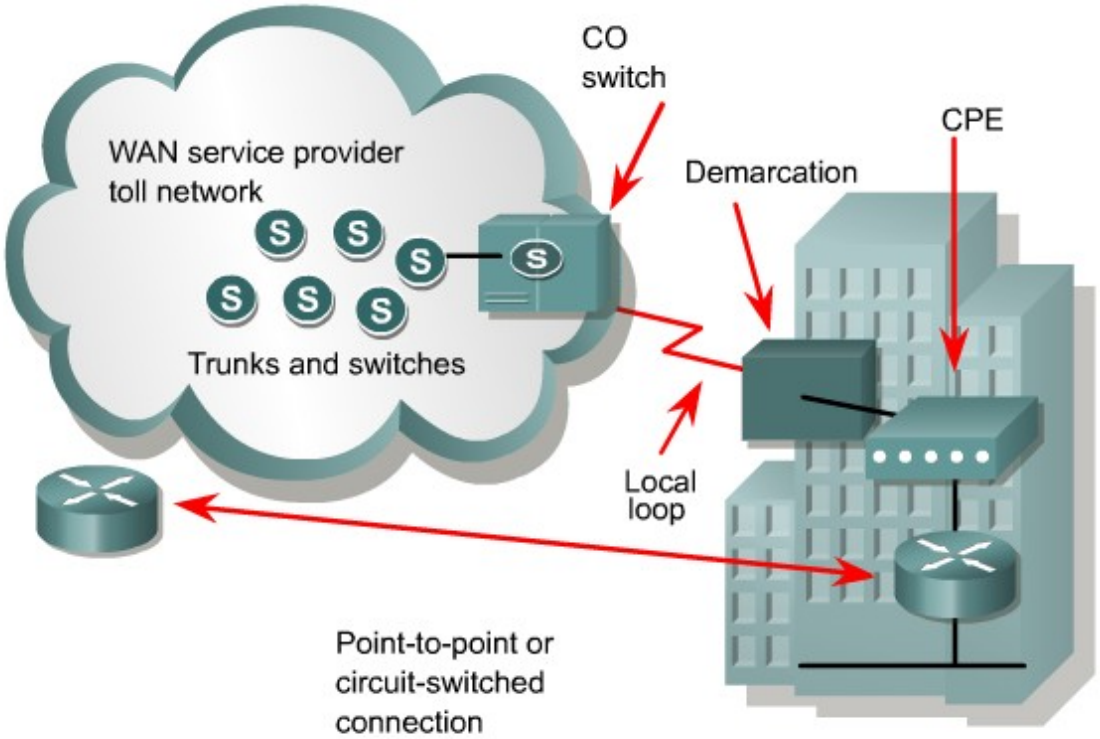
WAN Service Providers

Devices on the subscriber premises are called customer premises equipment (CPE).

The subscriber owns the CPE or leases the CPE from the service provider.

A copper or fiber cable connects the CPE to the service provider's nearest exchange or central office (CO).

This cabling is often called the local loop, or "last-mile".



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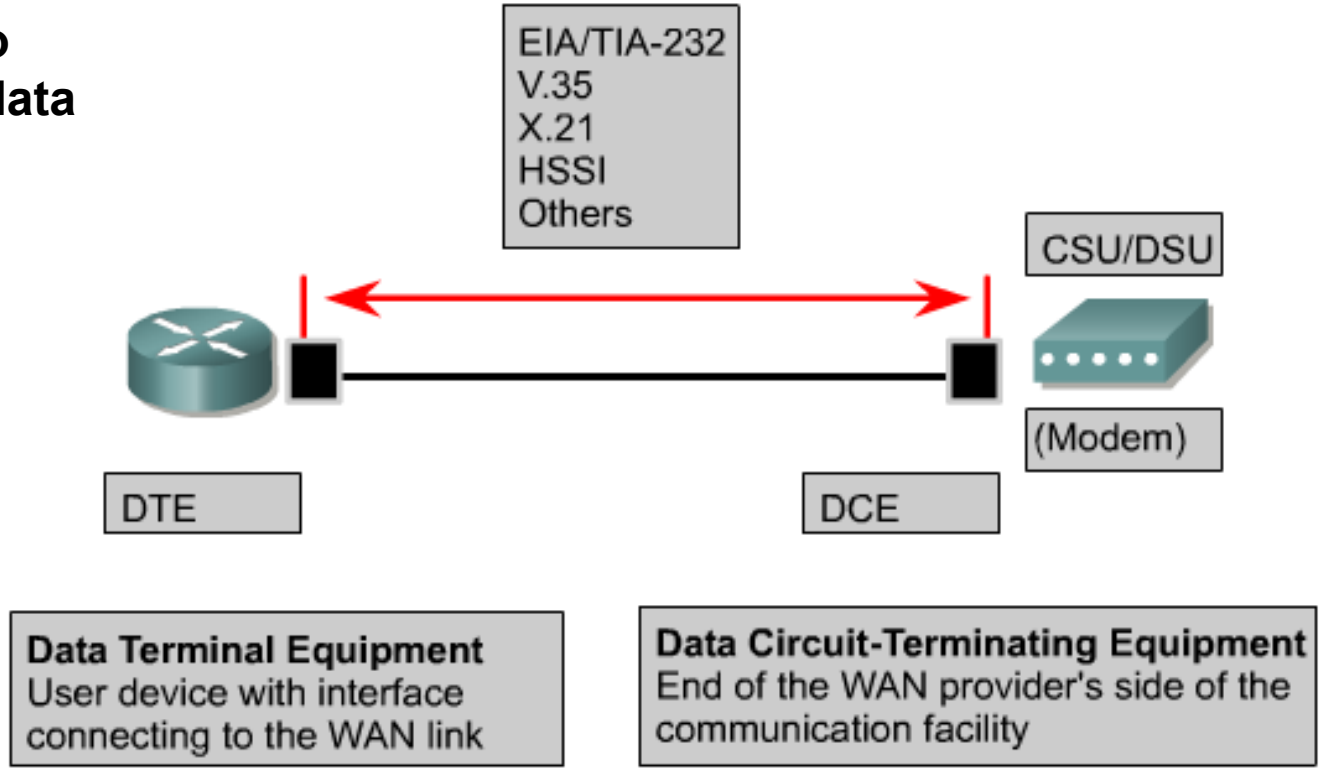
Physical Layer: WANs

Devices that put data on the local loop are called data circuit-terminating equipment, or data communications equipment (DCE).

The customer devices that pass the data to the DCE are called data terminal equipment (DTE).



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WAN Line Types and Bandwidth

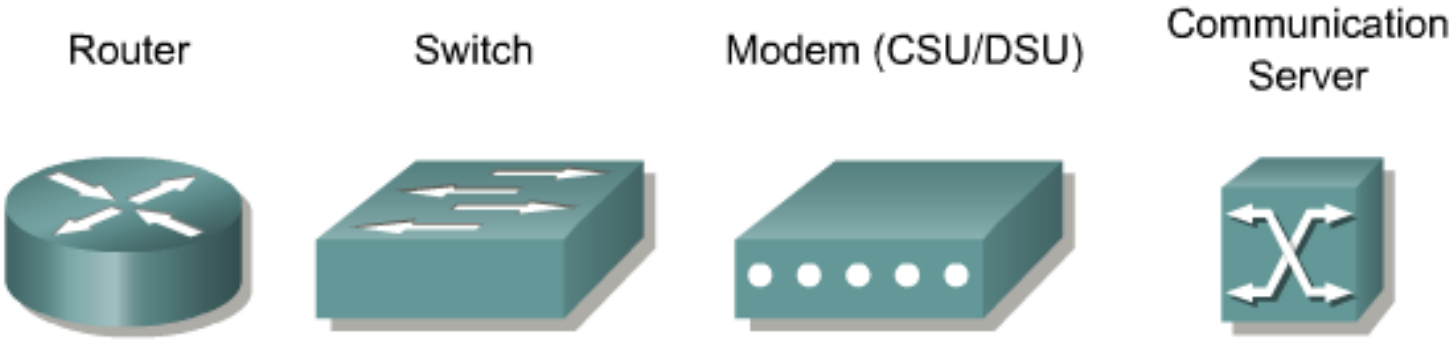
WAN links are provided at various speeds measured in bits per second (bps), kilobits per second (kbps or 1000 bps), megabits per second (Mbps or 1000 kbps) or gigabits per second (Gbps or 1000 Mbps).

Line Type	Signal Standard	Bit Rate Capacity
56	DS0	56 Kbps
64	DS0	64 Kbps
T1	DS1	51.84 Mbps
E1	ZM	2.048 Mbps
E3	M3	34.064 Mbps
J1	Y1	2.048 Mbps
T3	DS3	44.736 Mbps
OC-1	SONET	51.84 Mbps
OC-3	SONET	155.54 Mbps
OC-9	SONET	466.56 Mbps
OC-12	SONET	622.08 Mbps
OC-18	SONET	933.12 Mbps
OC-24	SONET	1244.16 Mbps
OC-36	SONET	1866.24 Mbps
OC-48	SONET	2488.32 Mbps

WAN Devices

WANs are groups of LANs connected together with communications links from a service provider.

Because the communications links cannot plug directly into the LAN, it is necessary to identify the various pieces of interfacing equipment.



WANs are designed to:

- Operate over large geographic area.
- Allow access over serial interfaces operating at lower speeds.
- Provide full-time and part-time connectivity.

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CSU/DSU

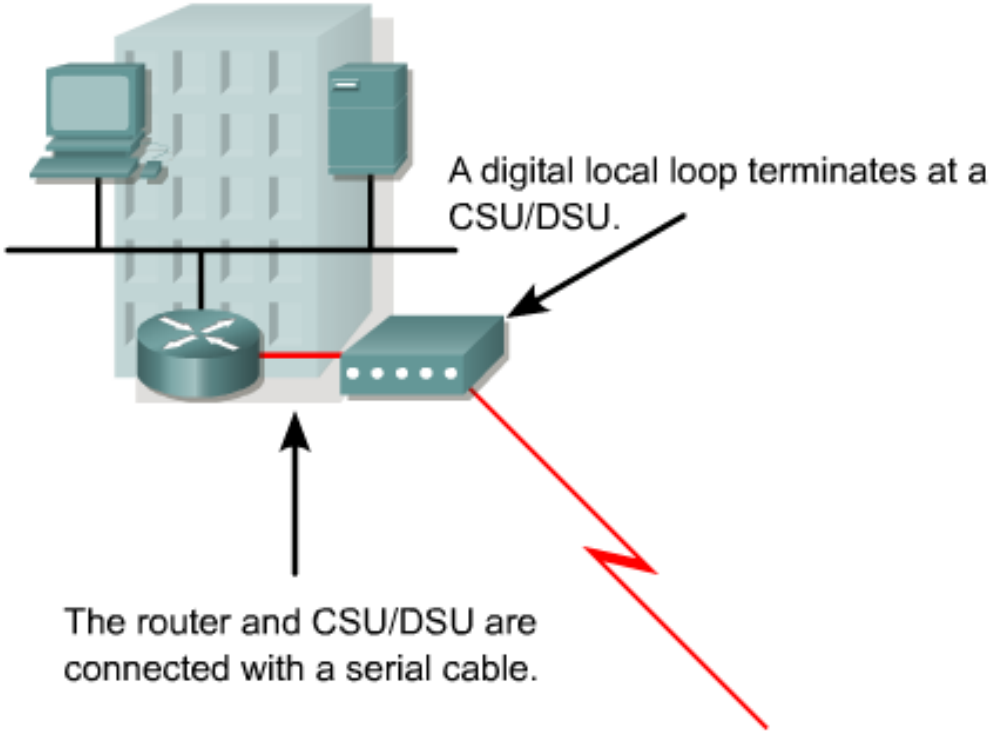
For digital lines, a channel service unit (CSU) and a data service unit (DSU) are required.

The two are often combined into a single piece of equipment, called the CSU/DSU.

The CSU/DSU may also be built into the interface card in the router.

CSU: Channel Service Unit, digital interface device that connects end-user equipment to the local digital telephone loop.

DSU: Data Service Unit, device used in digital transmission that adapts the physical interface on a DTE device to a transmission facility such as a T1 or an E1.



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Modem Transmission

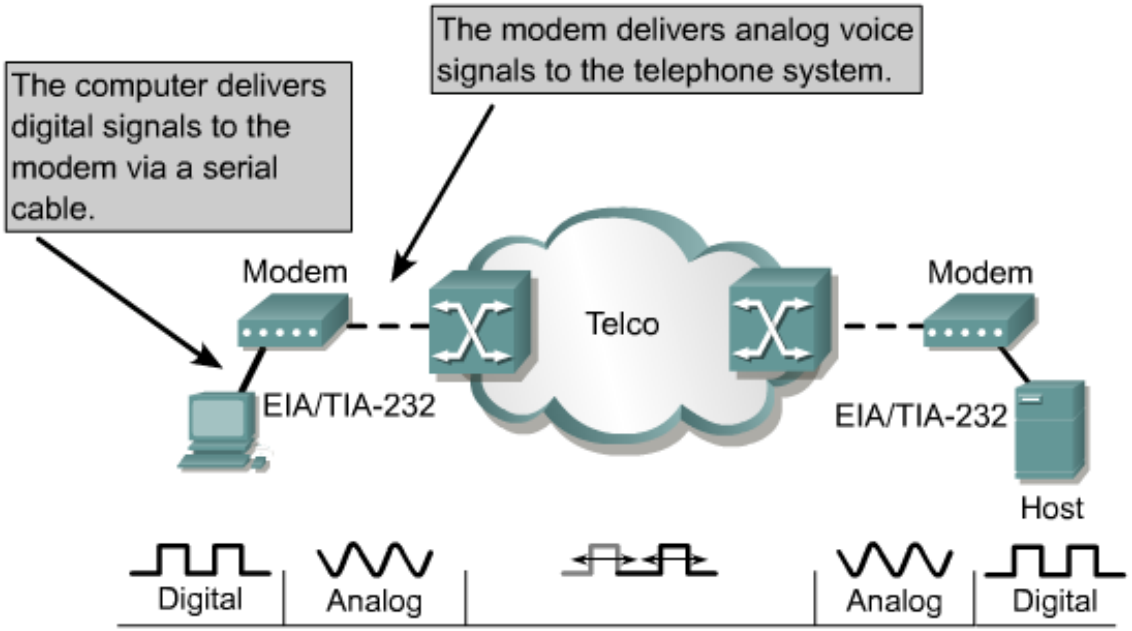
A modem is needed if the local loop is analog rather than digital.

Modems transmit data over voice-grade telephone lines by modulating and demodulating the signal.

The digital signals are superimposed on an analog voice signal that is modulated for transmission.

The modulated signal can be heard as a series of whistles by turning on the internal modem speaker.

At the receiving end the analog signals are returned to their digital form, or demodulated.



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WAN Standards

WANs use the OSI reference model, but focus mainly on Layer 1 and Layer 2.

WAN standards typically describe both physical layer delivery methods and data link layer requirements, including physical addressing, flow control, and encapsulation.

WAN standards are defined and managed by a number of recognized authorities.

Acronym	Organization
ITU-T (was CCITT)	ITU-T (was CCITT) International Telecommunication Union Telecommunication Standardization Sector, formerly the Consultative Committee for International Telegraph and Telephone
ISO	International Organization for Standardization
IETF	Internet Engineering Task Force
EIA	Electronic Industries Association
TIA	Telecommunications Industries Association

Standard	Description
ITU-T (was CCITT)	Allows signal speeds of up to 64 kbps on a 25 pin D connector over short distances. It was formerly known as RS-232. The ITU-T V.24 specification is effectively the same.
EIA/TIA 449/530	A faster (up to 2 Mbps) version of EIA/TIA 232. It uses a 36 pin D connector and is capable of longer cable runs. There are several versions. Also known as RS-422 and RS-423.
EIA/TIA 612/613	The High Speed Serial Interface (HSSI), which provides access to services at up to 52 Mbps on a 60 pin D connector.
V.35	An ITU-T standard for synchronous communications between a network access device and a packet network at speeds up to 48 kbps. It uses a 34 pin rectangular connector.
X.21	An ITU-T standard for synchronous digital communications. It uses a 15 pin D connector.

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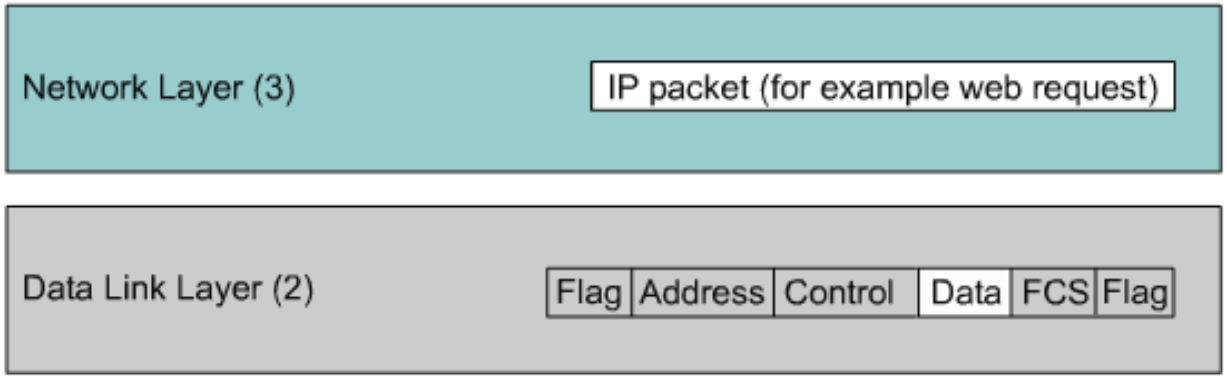
WAN Encapsulation

To ensure that the correct encapsulation protocol is used, the Layer 2 encapsulation type used for each router serial interface must be configured.

The choice of encapsulation protocols depends on the WAN technology and the equipment.

Most framing is based on the HDLC standard.

HDLC framing gives reliable delivery of data over unreliable lines and includes signal mechanisms for flow and error control.



Network data is encapsulated in an HDLC frame.

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WAN Data-Link Protocols

Both PPP and the Cisco version of HDLC have an extra field in the header to identify the network layer protocol of the encapsulated data.

Protocol	Usage
Link Access Protocol Balanced (LAPB)	X.25
Link Access Protocol D Channel (LAPD)	ISDN D channel
Link Access Protocol Frame (LAPF)	Frame Relay
High-Level Data Link Control (HDLC)	Cisco's implementation has an extra header field
Point-to-Point Protocol (PPP)	Dialup connections

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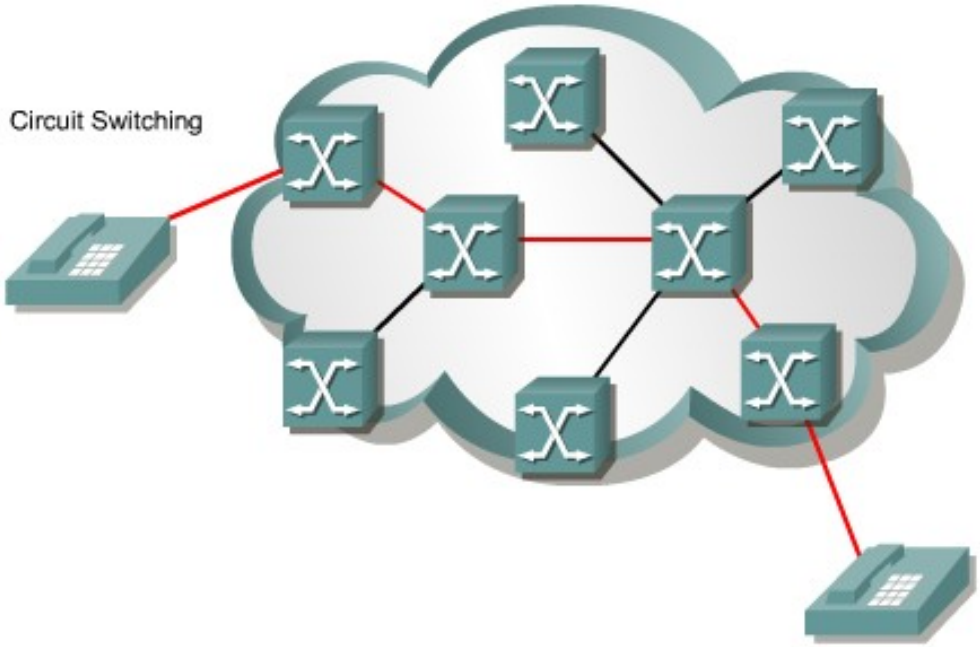
Circuit Switching

When a subscriber makes a telephone call, the dialed number is used to set switches in the exchanges along the route of the call so that there is a continuous circuit from the originating caller to that of the called party.

Because of the switching operation used to establish the circuit, the telephone system is called a circuit-switched network.

If the telephones are replaced with modems, then the switched circuit is able to carry computer data.



Dialing sets up a physical circuit through the system.

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Packet Switching

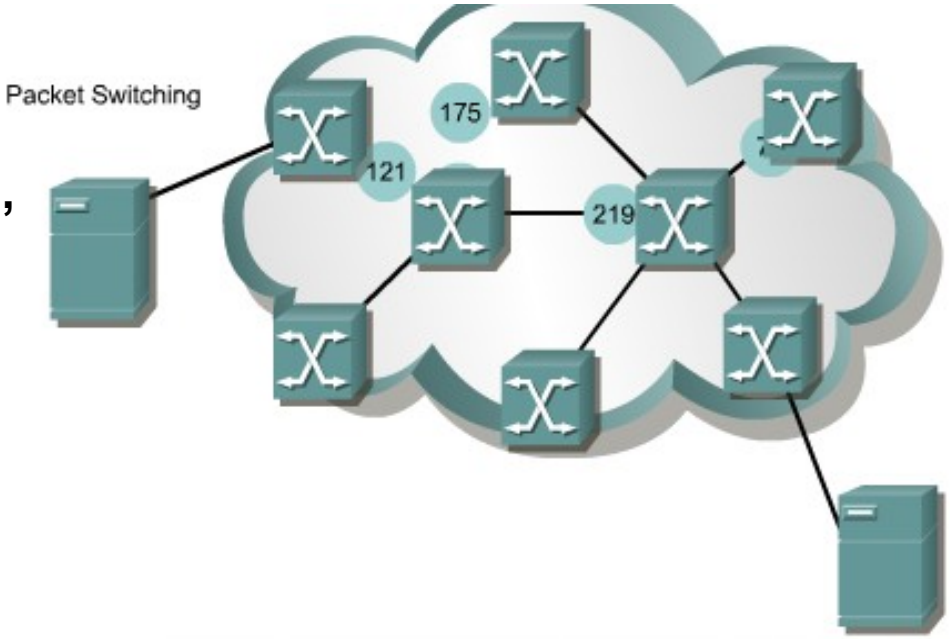
An alternative to Circuit switches is to allocate the capacity to the traffic only when it is needed, and share the available capacity between many users. With a circuit-switched connection, the data bits put on the circuit are automatically delivered to the far end because the circuit is already established.

If the circuit is to be shared, there must be some mechanism to label the bits so that the system knows where to deliver them.

It is difficult to label individual bits, therefore they are gathered into groups called cells, frames, or packets.

The packet passes from exchange to exchange for delivery through the provider network.

Networks that implement this system are called packet-switched networks.



Labeled data is passed from switch to switch. It may have to wait its turn on a link.

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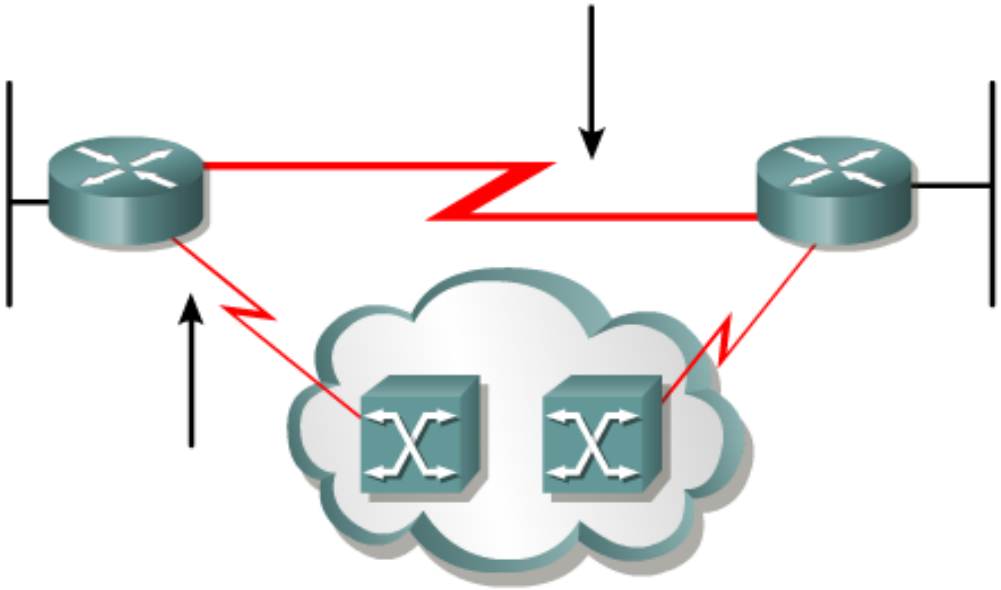
WAN Link Options

To connect to a packet-switched network, a subscriber needs a local loop to the nearest location where the provider makes the service available.

This is called the point-of-presence (POP) of the service.

Normally this will be a dedicated leased line.

This line will be much shorter than a leased line directly connected to the subscriber locations, and often carries several VCs.



LANs can be connected with a long leased line or with shorter leased lines to a packet switched network. The packet network does the long haul.

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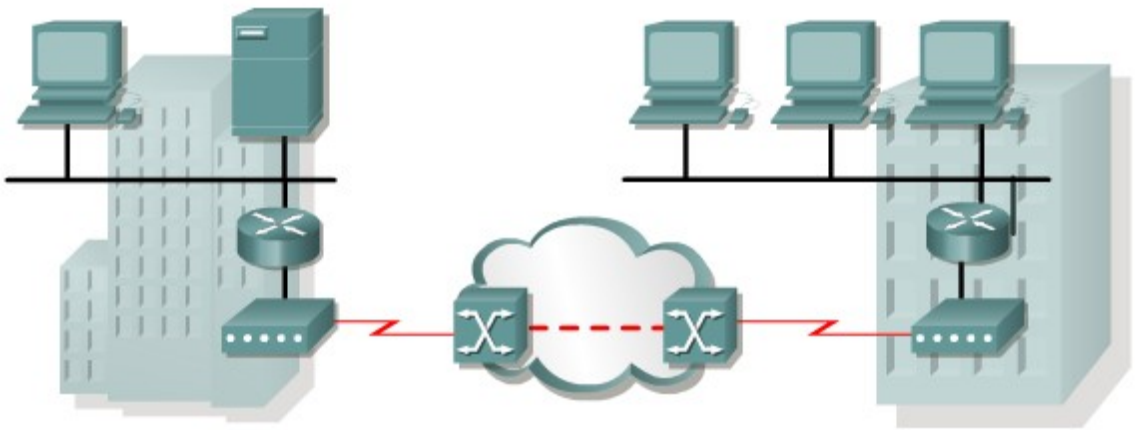
Analog Dialup

When intermittent, low-volume data transfers are needed, modems and analog dialed telephone lines provide low capacity and dedicated switched connections.

The advantages of modem and analog lines are simplicity, availability, and low implementation cost.

The disadvantages are the low data rates and a relatively long connection time.

The dedicated circuit provided by dialup will have little delay or jitter for point-to-point traffic, but voice or video traffic will not operate adequately at relatively low bit rates.



WAN built with intermittent connections using a modem and the voice telephone network.

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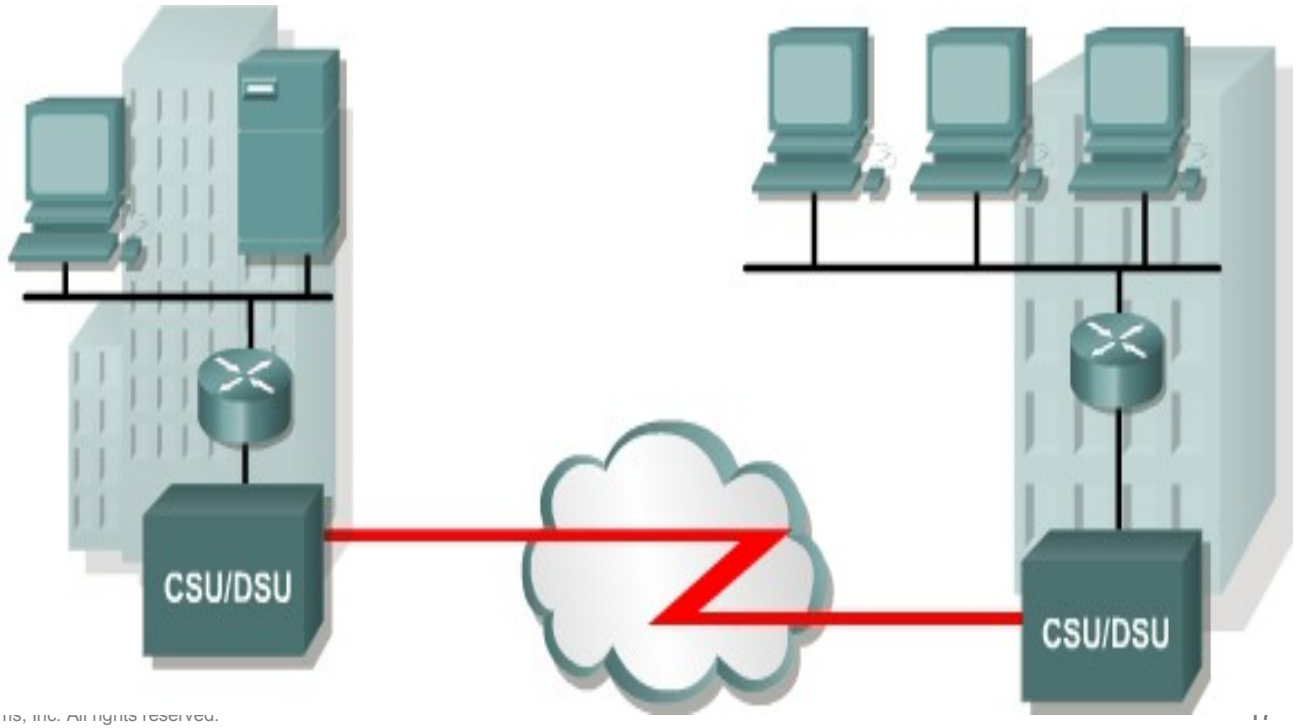
Leased Line

Leased lines are not only used to provide direct point-to-point connections between Enterprise LANS, they can also be used to connect individual branches to a packet switched network.

- **Point-to-point lines are usually leased from a carrier and are called leased lines.**
- **Leased lines are available in different capacities.**
- **The cost of leased-line solutions can become significant when they are used to connect many sites.**



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WAN with X.25

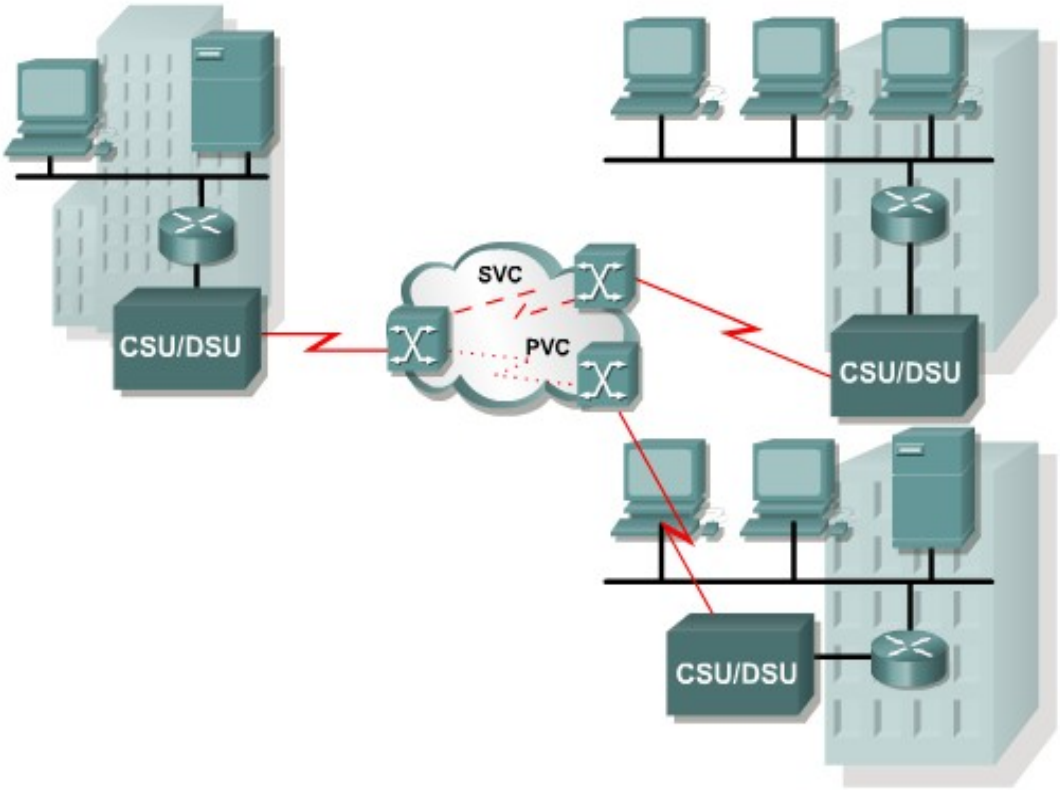
X.25 provides a low bit rate, shared-variable capacity that may either be switched or permanent.

X.25 can be very cost effective because tariffs are based on the amount of data delivered rather than connection time or distance.

Data can be delivered at any rate up to the connection capacity.

X.25 networks are usually low capacity, with a maximum of 48 kbps.

In addition, the data packets are subject to the delays typical of shared networks.



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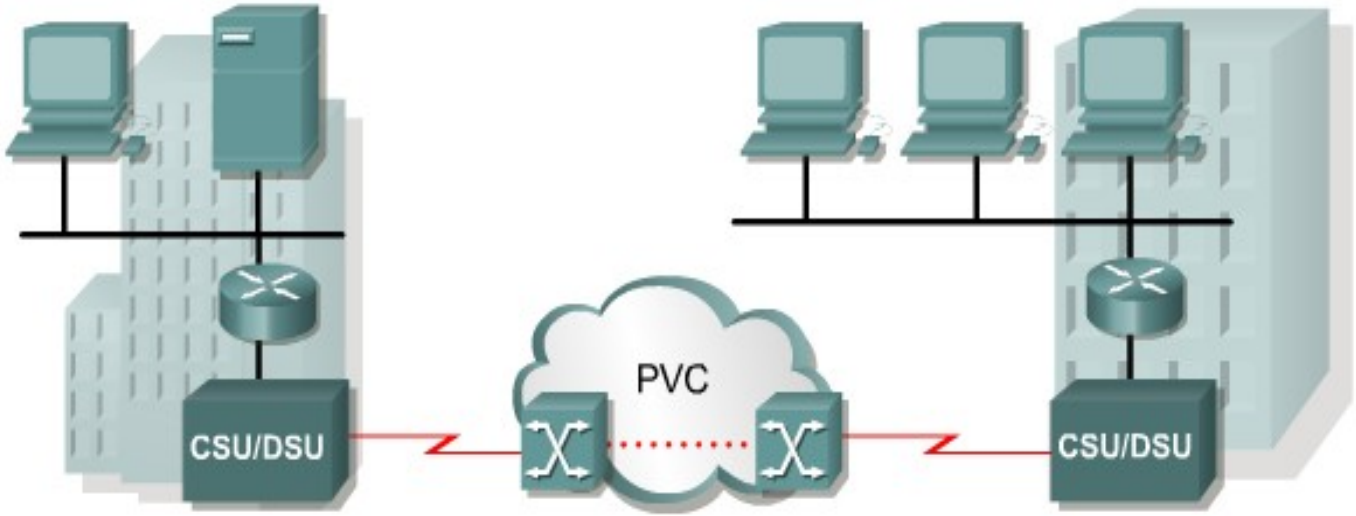
Frame Relay

Most Frame Relay connections are based on PVCs rather than SVCs.

It implements no error or flow control.

Frame Relay provides permanent shared medium bandwidth connectivity that carries both voice and data traffic.



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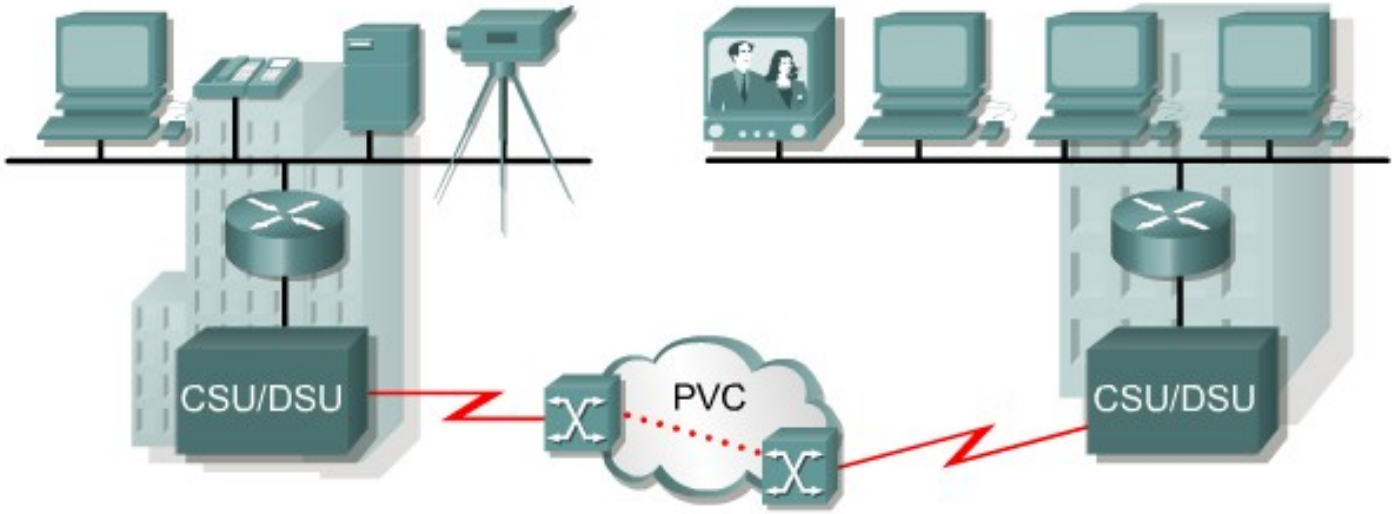
ATM

Asynchronous Transfer Mode (ATM) is a technology capable of transferring voice, video, and data through private and public networks. It is built on a cell based architecture rather than on a frame-based architecture.



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DSL

Digital Subscriber Line (DSL) technology is a broadband technology that uses existing twisted-pair telephone lines to transport high-bandwidth data to service subscribers.

DSL technology allows the local loop line to be used for normal telephone voice connection and an always-on connection for instant network connectivity.

Service	Download	Upload
ADSL	1.5 M - 8.192 M	16 K - 640 K
SDSL	1.544 M - 2.048 M	1.544 M - 2.048 M
HDSL	1.544 M - 2.048 M	1.544 M - 2.048 M
IDSL	144 K	144 K
RADSL	64 K - 8.192 M	16 M - 768 M
CDSL	1 M	16 K -160 K

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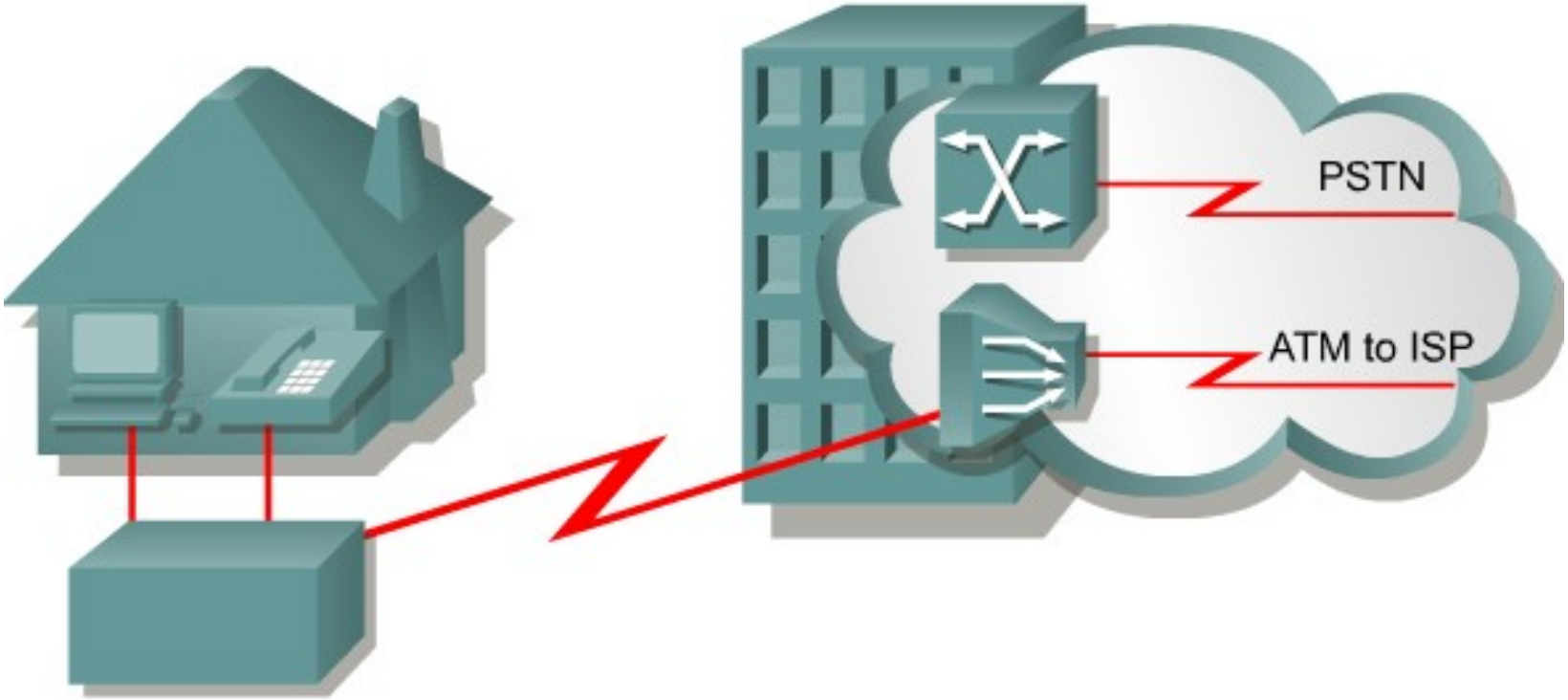
ADSL Technology



Local loop connects splitter in house to DSLAM in central office.
Voice and data use separate frequency ranges.

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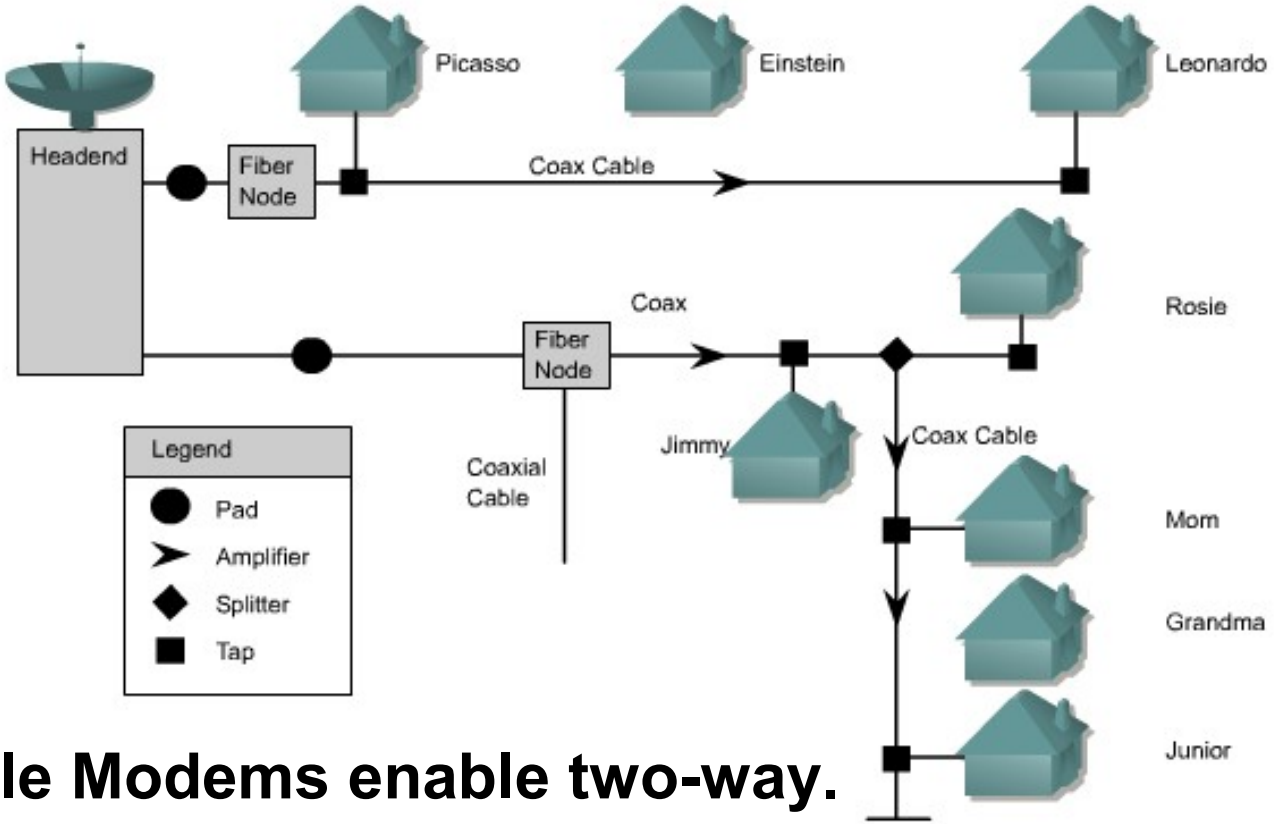
Cable Modem



- **Enhanced Cable Modems enable two-way.**
- **High speed data transmissions using the same coaxial lines that transmit cable television.**

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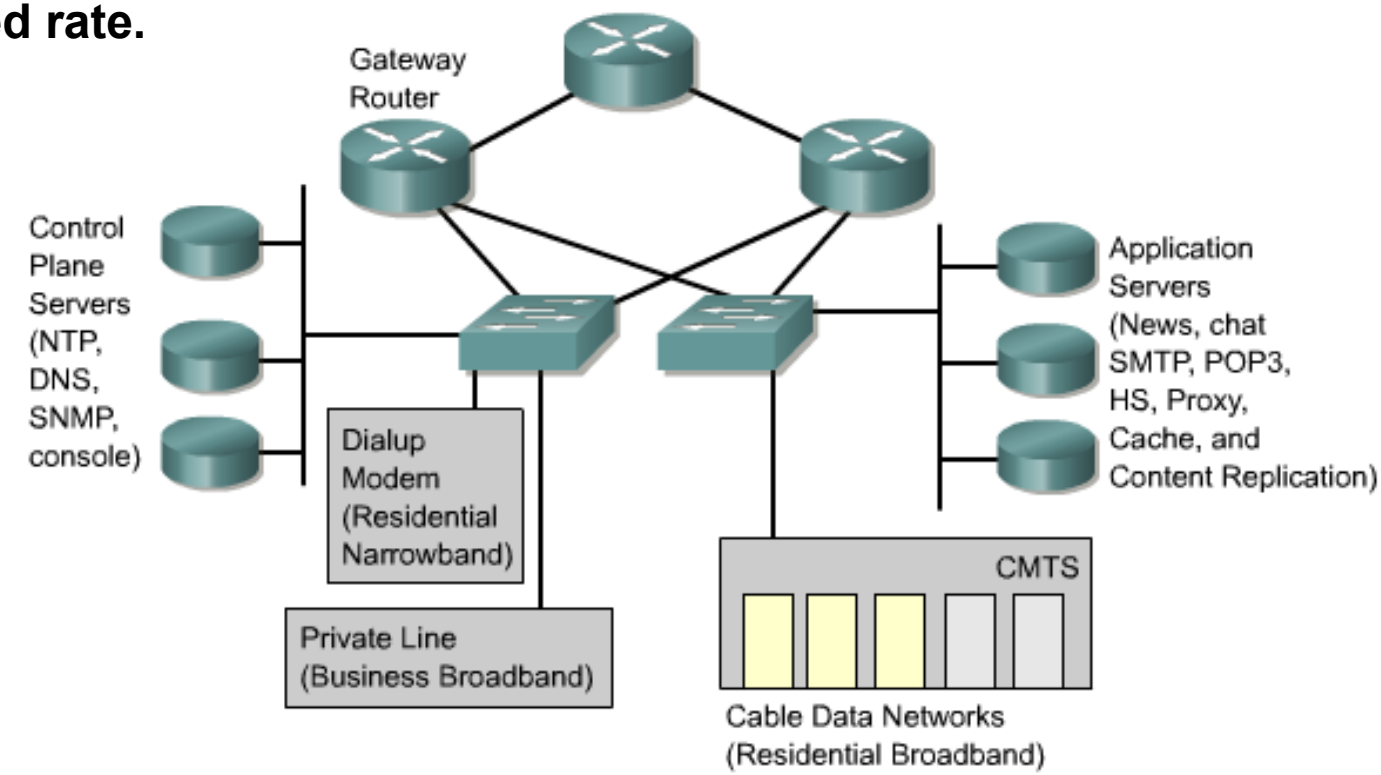
Cable Data Network Architecture

Cable modem subscribers must use the ISP associated with the service provider.

All the local subscribers share the same cable bandwidth.

As more users join the service, available bandwidth may be below the expected rate.



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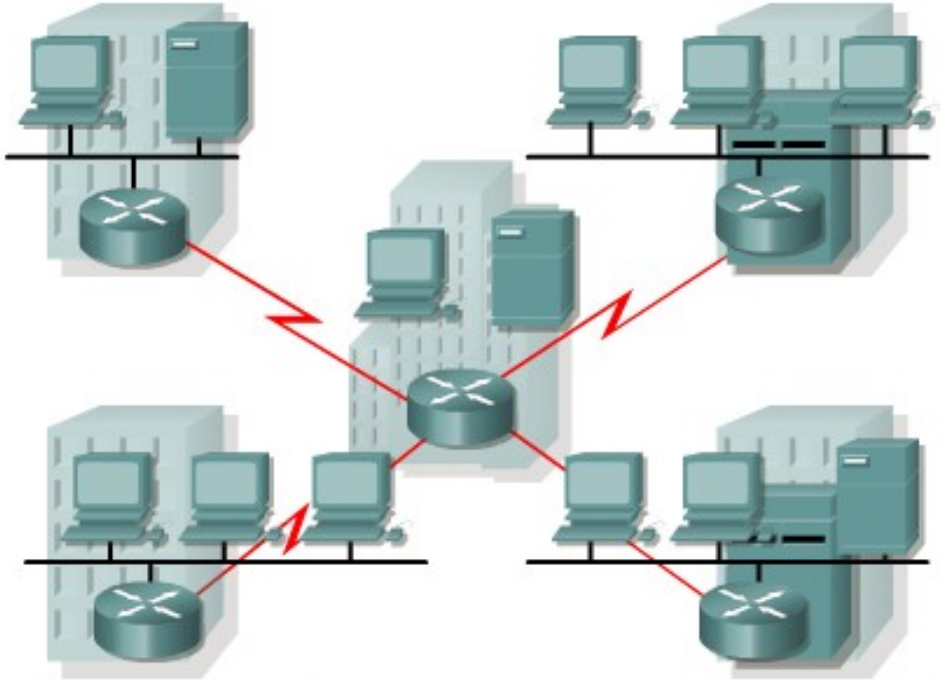
Modern WAN

WANs carry a variety of traffic types such as data, voice, and video.

The design selected must provide adequate capacity and transit times to meet the requirements of the enterprise.

Among other specifications, the design must consider the topology of the connections between the various sites, the nature of those connections, and bandwidth capacity.



LANs separated by distance are linked by data communications lines and routers.

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Comparing WAN Traffic Types

Traffic	Latency	Jitter	Bandwidth
Voice	Low	Low	Medium
Transaction data (for example, SNA)	Medium	Medium	Medium
Messaging (e-mail)	High	High	High
File transfer	High	High	High
Batch data	High	High	High
Network management	High	High	Low
Videoconferencing	Low	Low	High

Some WAN traffic types with tolerance to latency and jitter, along with bandwidth requirements.

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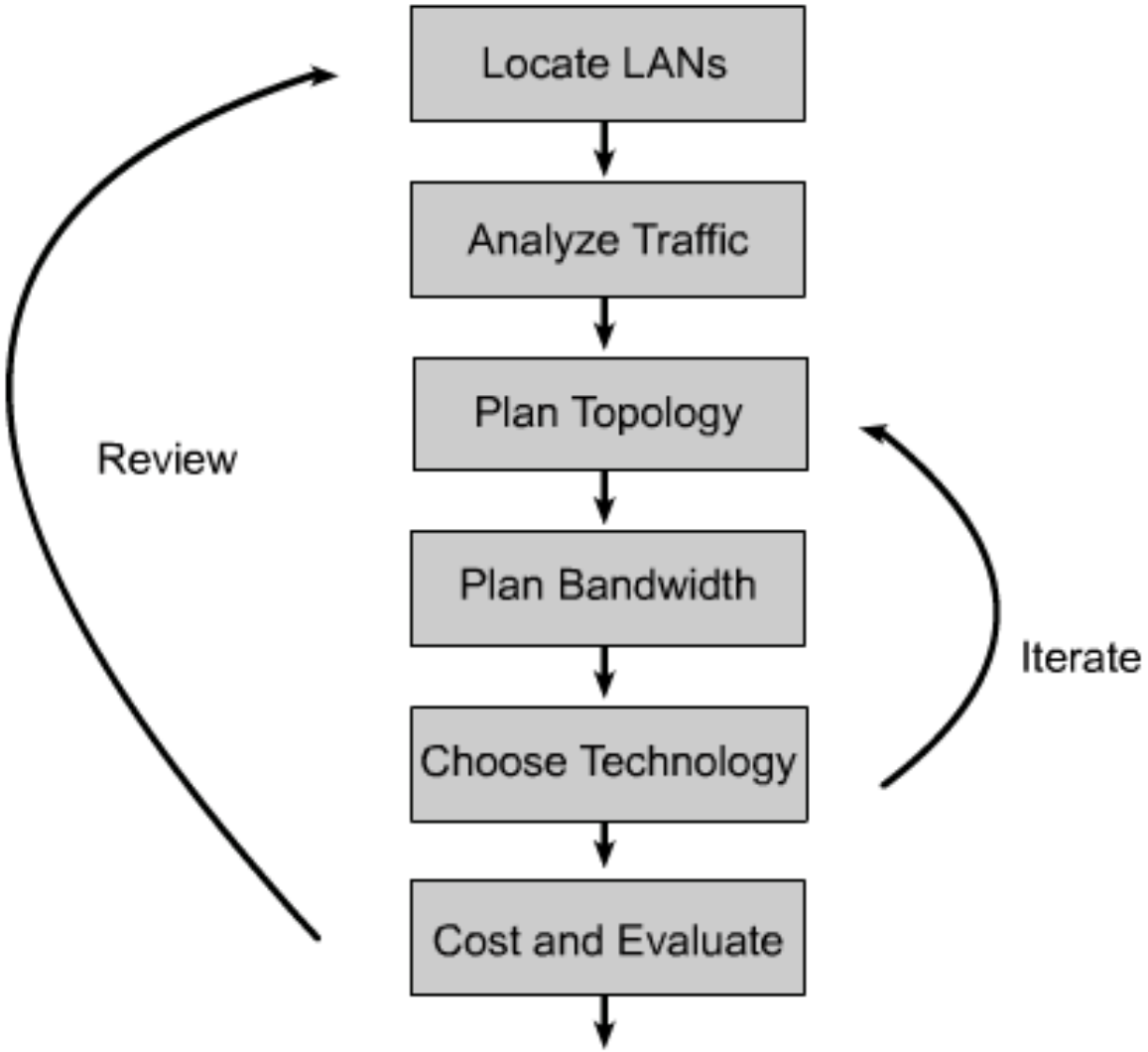
Steps In WAN Design

In practice, following the steps shown is seldom a linear process.

Several modifications may be necessary before a design is finalized.

Continued monitoring and re-evaluation are also required after installation of the WAN to maintain optimal performance.



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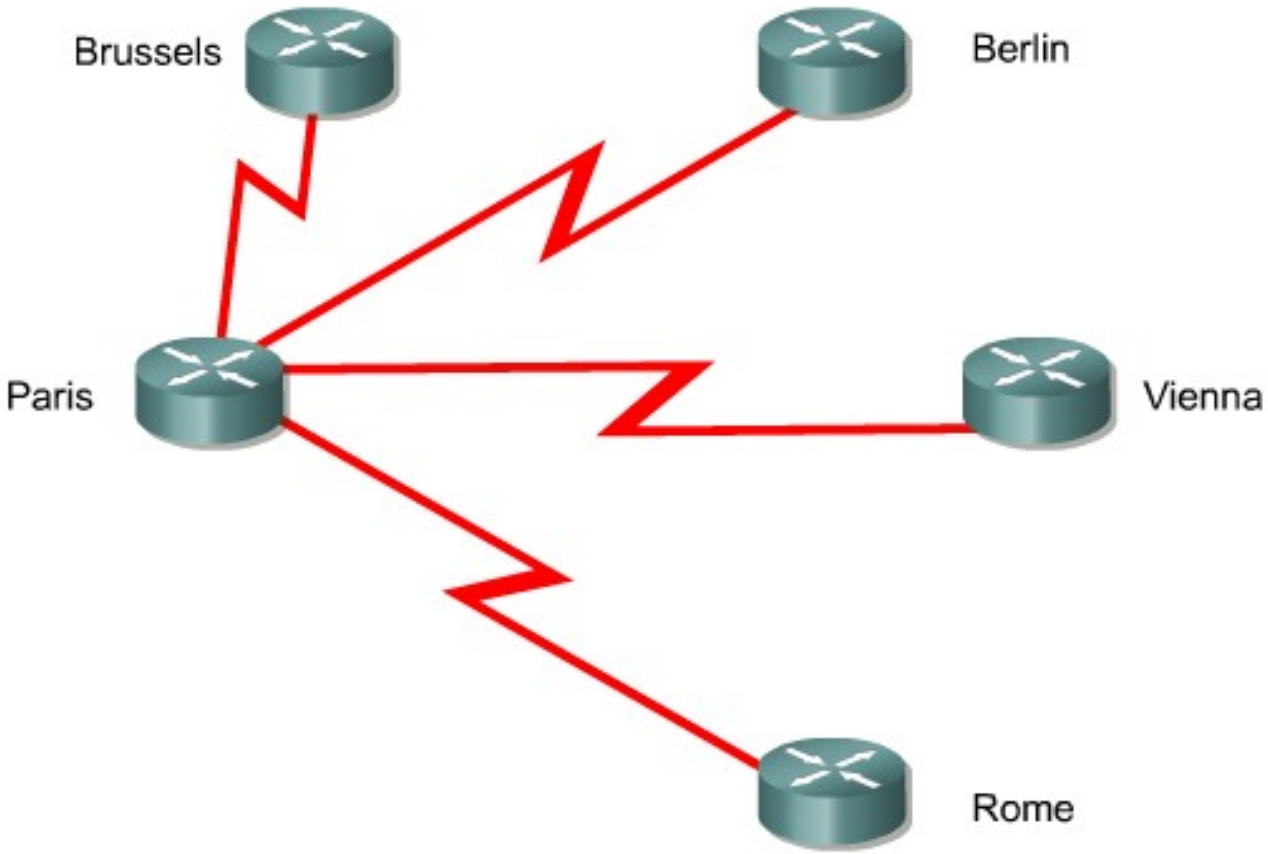
Star Topology

Many WANs use a star topology.

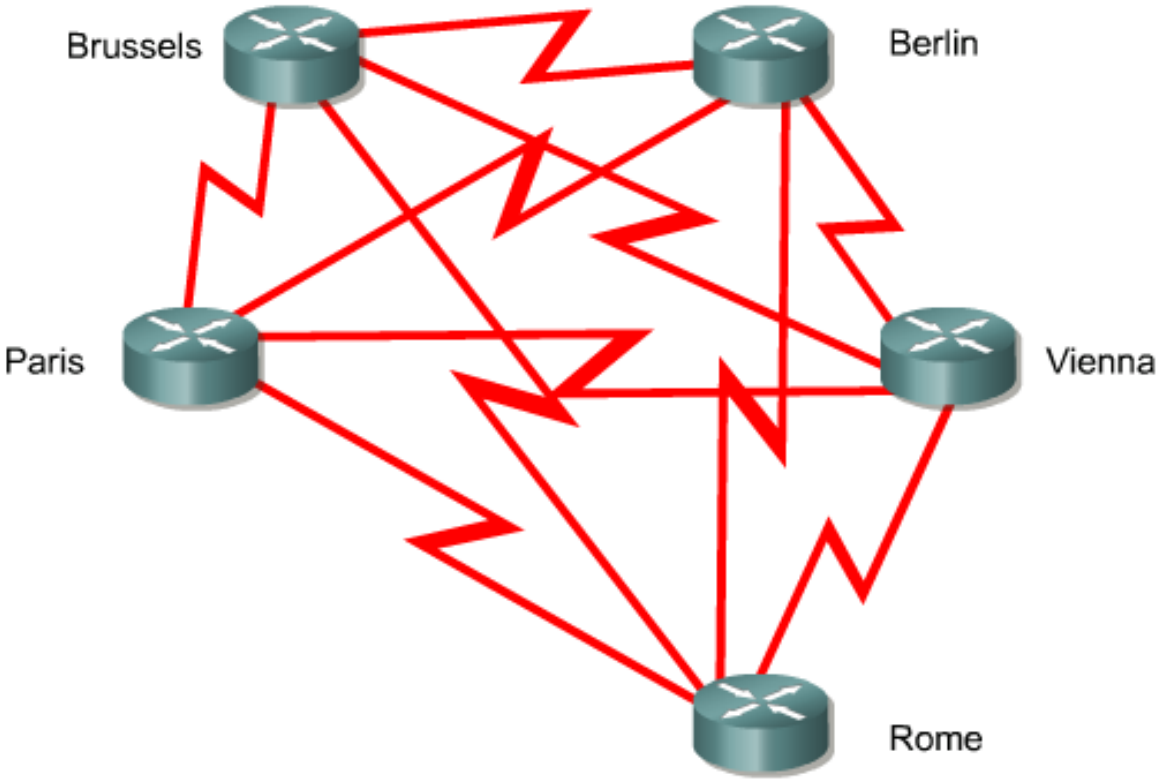
As the enterprise grows and new branches are added, the branches are connected back to the head office, producing a traditional star topology.



Full-Mesh Topology

Star end-points are sometimes cross-connected, creating a mesh or partial mesh topology.

This provides for many possible combinations for interconnections.



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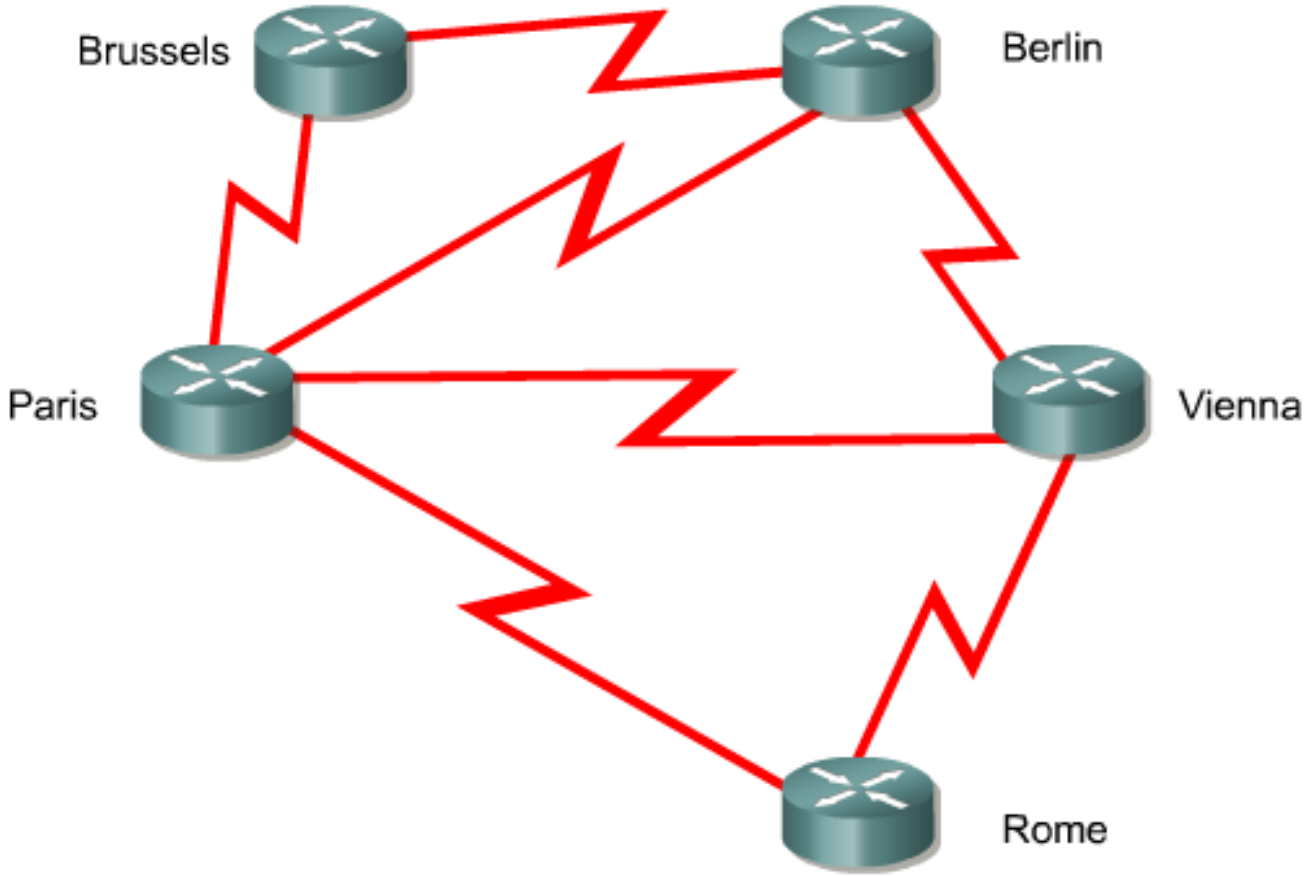
Partial-Mesh Topology

When designing, re-evaluating, or modifying a WAN, a topology that meets the design requirements must be selected.



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Three-Layer Design Model

The links connecting the various sites in an area that provide access to the enterprise network are called the access links or access layer of the WAN.

Traffic between areas is distributed by the distribution links, and is moved onto the core links for transfer to other regions, when necessary.

This hierarchy is often useful when the network traffic mirrors the enterprise branch structure and is divided into regions, areas, and branches.

It is also useful when there is a central service to which all branches must have access, but traffic levels are insufficient to justify direct connection of a branch to the service.

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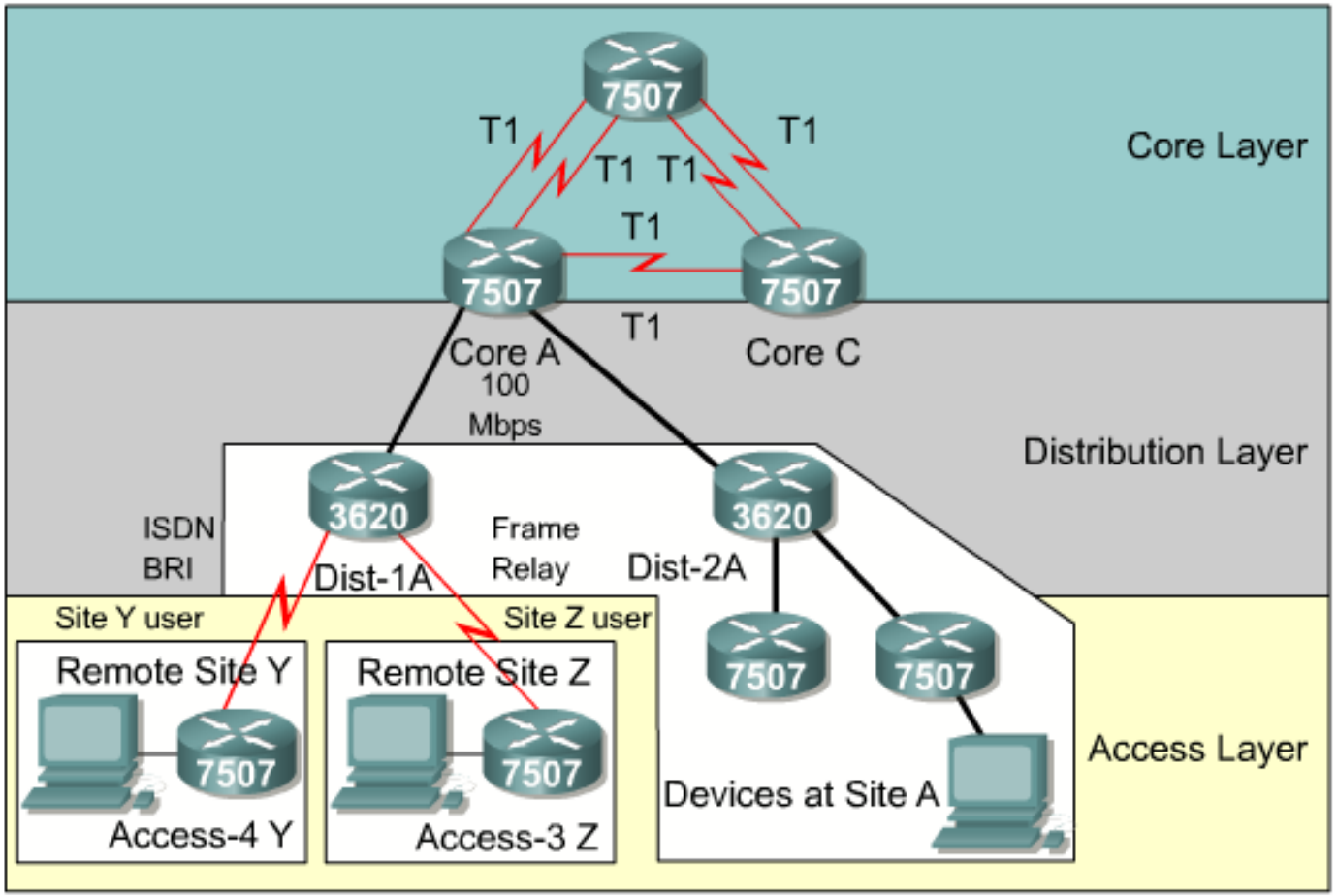
Internet for WAN Connectivity

Many networks do not require the complexity of a full three-layer hierarchy.

Simpler hierarchies may be used.



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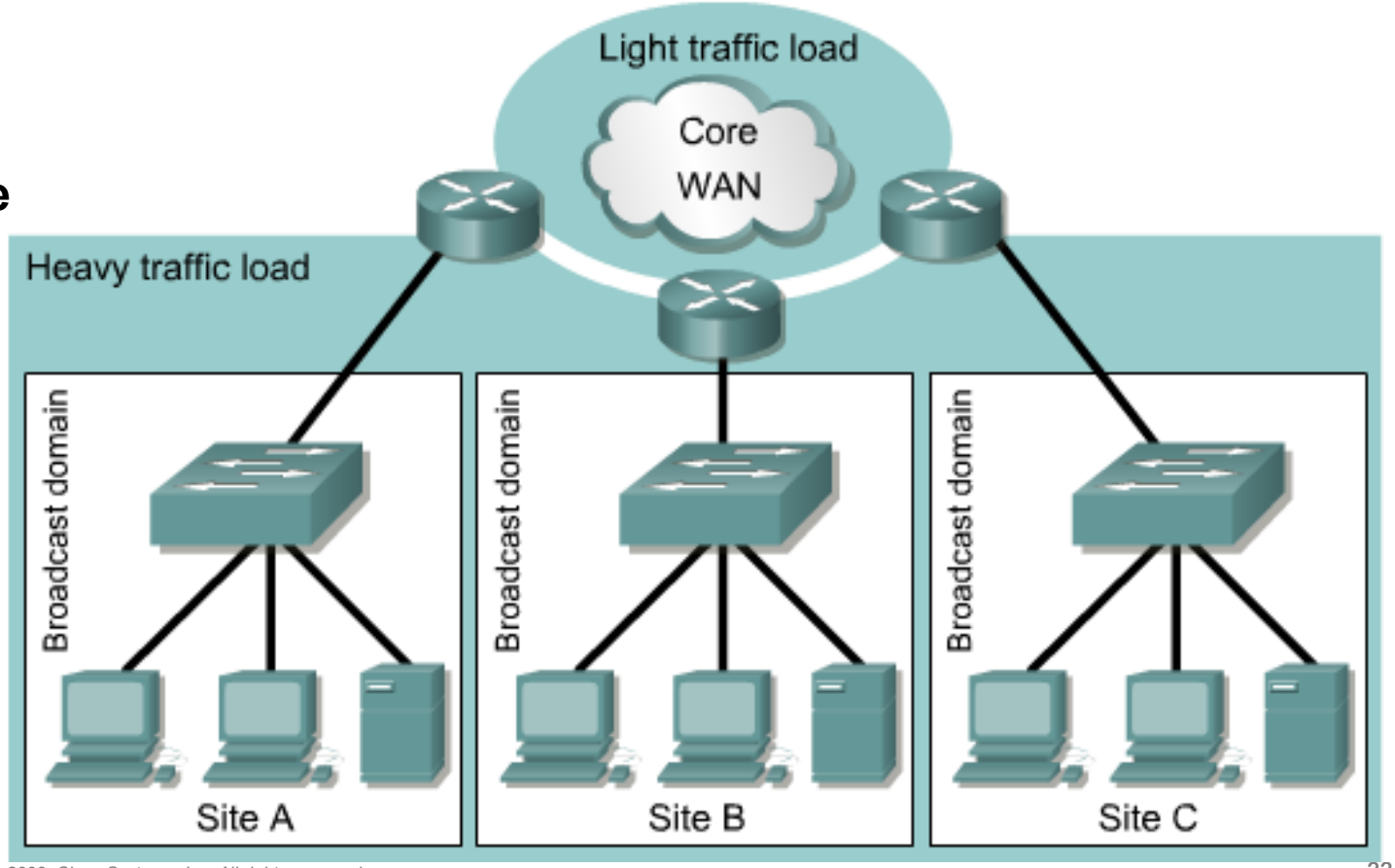
One-Layer Hierarchy

An enterprise with several relatively small branches that require minimal inter-branch traffic may choose a one-layer design.

Historically this has not been popular because of the length of the leased lines.

Frame Relay, where charges are not distance related, is now making this a feasible design solution.



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